Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains

Abstract: *The rapid integration of Large Language Models (LLMs) into scientific workflows necessitates rigorous evaluation of their factual accuracy. While LLMs excel in open-ended text generation, they are prone to producing hallucinations—plausible but incorrect statements—which pose critical risks in high-stakes domains such as medicine, biology, and the physical sciences. This paper provides a comprehensive review of the current paradigms for benchmarking factual accuracy in LLMs across scientific domains. We examine foundational datasets, including SciFact and PubMedQA, and explore the mechanisms by which models mimic human falsehoods, as demonstrated by the TruthfulQA benchmark. Furthermore, we analyze recent advancements in automated factuality evaluation, such as FActScore and SelfCheckGPT, alongside the role of retrieval-augmented generation and search-engine augmentation in mitigating factual drift over time. Finally, this paper highlights critical research gaps and proposes future directions for developing dynamic, domain-specific evaluation frameworks that can ensure the reliability of LLMs in scientific research.*

Keywords: Large Language Models, Factual Accuracy, Hallucination Detection, Scientific Domains, Benchmarking, Fact-Checking.

1. Introduction

Large language models have revolutionized natural language processing (NLP) and are increasingly applied to scientific research, literature summarization, and clinical decision-making. Despite their remarkable fluency, a persistent challenge is their tendency to "hallucinate"—generating text that is grammatically correct but factually incorrect or unsupported by empirical evidence. In scientific domains, where precision and verifiable truth are paramount, the deployment of unchecked LLMs presents significant risks. Consequently, benchmarking the factual accuracy of these models has emerged as a critical sub-field in AI research. This paper systematically reviews the methodologies, datasets, and algorithmic frameworks designed to measure and improve the factuality of LLMs in scientific contexts. By examining both static benchmarks and dynamic evaluation pipelines, we aim to provide a holistic overview of how the AI community is addressing the challenge of scientific hallucination.

2. Background and Literature Review

The foundation of benchmarking LLM factuality lies in the development of specialized datasets that test reasoning and knowledge retrieval. Early efforts focused on specific scientific niches. For instance, in the biomedical domain, researchers introduced PubMedQA, a novel question-answering dataset collected directly from PubMed abstracts. PubMedQA requires models to answer research questions with yes, no, or maybe, forcing them to reason over quantitative and technical biomedical texts.

Similarly, the verification of scientific claims was formalized through the SciFact dataset. SciFact contains expert-written scientific claims paired with evidence-containing abstracts, allowing researchers to evaluate whether an AI system can correctly identify if a given abstract supports or refutes a claim. These domain-specific benchmarks highlighted that standard language models, primarily trained on general web corpora, often struggle with specialized scientific reasoning.

Beyond domain-specific datasets, the alignment of models with human truthfulness became a central focus. The seminal work on InstructGPT demonstrated that training language models to follow instructions using human feedback (RLHF) significantly improves their helpfulness and reduces overt toxicity, though challenges with subtle hallucinations remain. To specifically measure how models adopt human misconceptions, the TruthfulQA benchmark was developed, comprising adversarial questions designed to trigger false beliefs. The creators found that, paradoxically, larger language models are sometimes less truthful because they become better at imitating widespread human falsehoods present in the training data.

3. Main Technical Discussion

Evaluating factual accuracy in scientific domains requires moving beyond simple multiple-choice questions to analyzing long-form, open-ended generation. Traditional metrics like BLEU or ROUGE fail to capture factual consistency. To address this, pipelines like Multi-FAct have been developed, extending the FActScore framework to evaluate the factual accuracy of large language models across multiple languages and regional contexts, demonstrating that complex text can be broken down into atomic facts and verified against external knowledge bases.

Furthermore, the sheer scale of LLM outputs necessitates automated evaluation systems. The HaluEval benchmark was introduced as a large-scale hallucination evaluation collection, utilizing a "sampling-then-filtering" framework to generate and annotate hallucinated samples. This allows researchers to test whether LLMs can successfully recognize hallucinations within text, a capability that is critical if models are to act as automated scientific assistants.

Another technical approach involves manipulating the prompt structure to elicit better reasoning. Chain-of-thought prompting significantly improves the ability of large models to perform complex reasoning tasks by requiring them to output intermediate steps before arriving at a final answer. In scientific contexts, this step-by-step reasoning allows researchers to verify the logic leading to a conclusion, rather than treating the model as an opaque oracle.

4. Current Approaches and Developments

Current approaches to improving and benchmarking factuality increasingly rely on external knowledge retrieval. Models are being augmented with search engines or pre-trained with retrieval mechanisms. For example, recent studies have comprehensively explored whether autoregressive language models should be pretrained with retrieval, concluding that doing so can moderately improve factual accuracy and reduce repetition compared to standard, closed-book models.

When external databases are unavailable, zero-resource hallucination detection methods are employed. SelfCheckGPT uses a sampling-based approach to fact-check black-box model responses. The underlying premise is that if a model possesses genuine knowledge, multiple stochastically sampled responses will remain consistent; conversely, hallucinated facts will diverge and contradict one another.

To combat the issue of static knowledge cut-offs, dynamic benchmarks are actively being developed. FreshQA tests a model's ability to answer questions requiring up-to-date world knowledge and fast-changing facts. The findings demonstrate that even advanced models struggle with false premises unless augmented by real-time search engine data through techniques like FreshPrompt.

5. Research Challenges, Gaps, and Conclusion

Despite these advancements, several limitations persist. Scientific knowledge is highly specialized and rapidly evolving. Static benchmarks risk becoming outdated or contaminated if their contents are inadvertently included in the massive pre-training corpora of newer LLMs. Future research must bridge the gap between static evaluation and the dynamic nature of scientific discovery. There is a pressing need for continuously updating benchmarks that automatically source new claims from recent literature. Benchmarking the factual accuracy of LLMs across scientific domains is a multifaceted challenge that requires a combination of high-quality datasets, fine-grained evaluation metrics, and advanced prompting techniques. The integration of real-time retrieval systems will be essential to harnessing the power of LLMs safely and effectively in academic research.

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